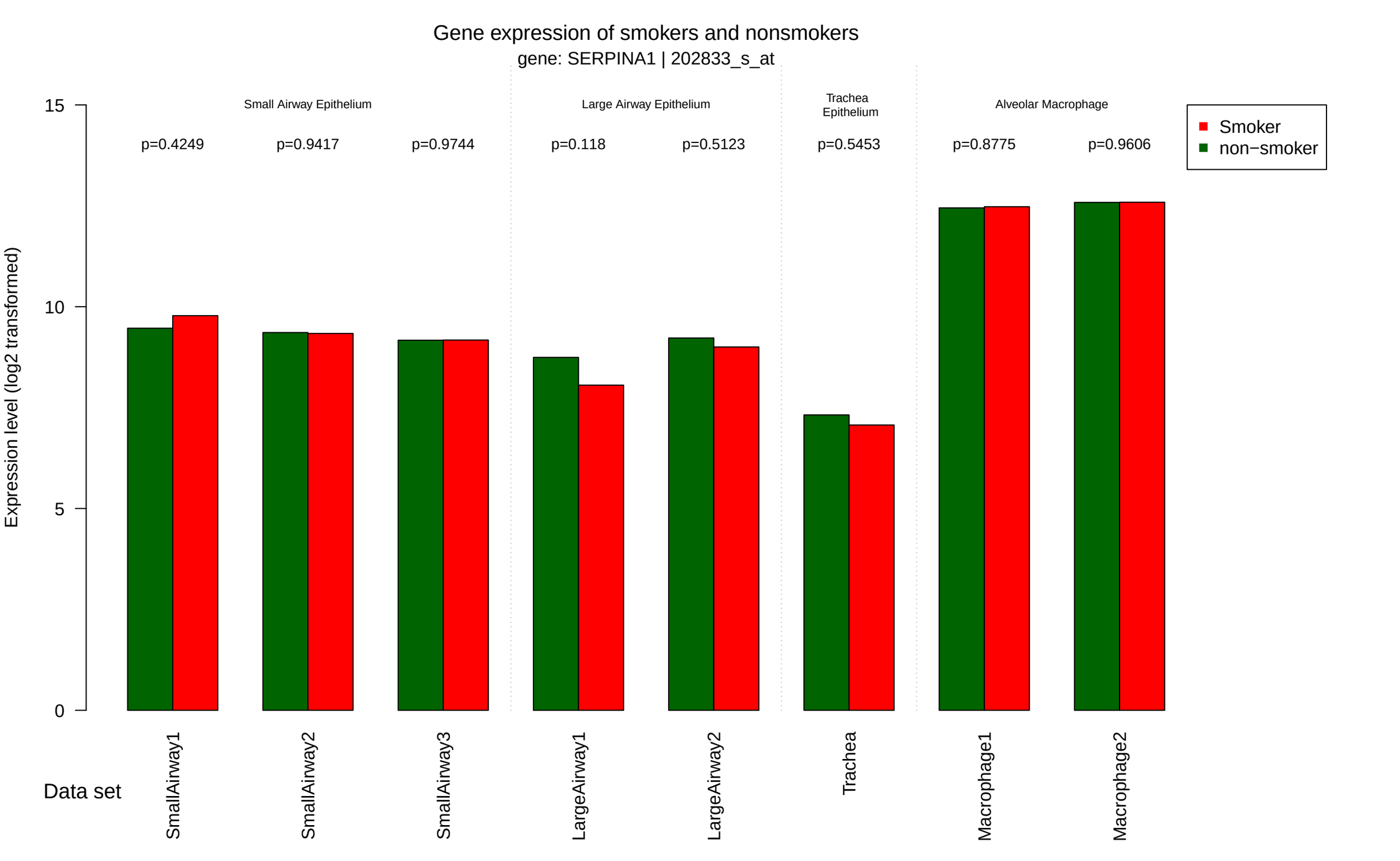
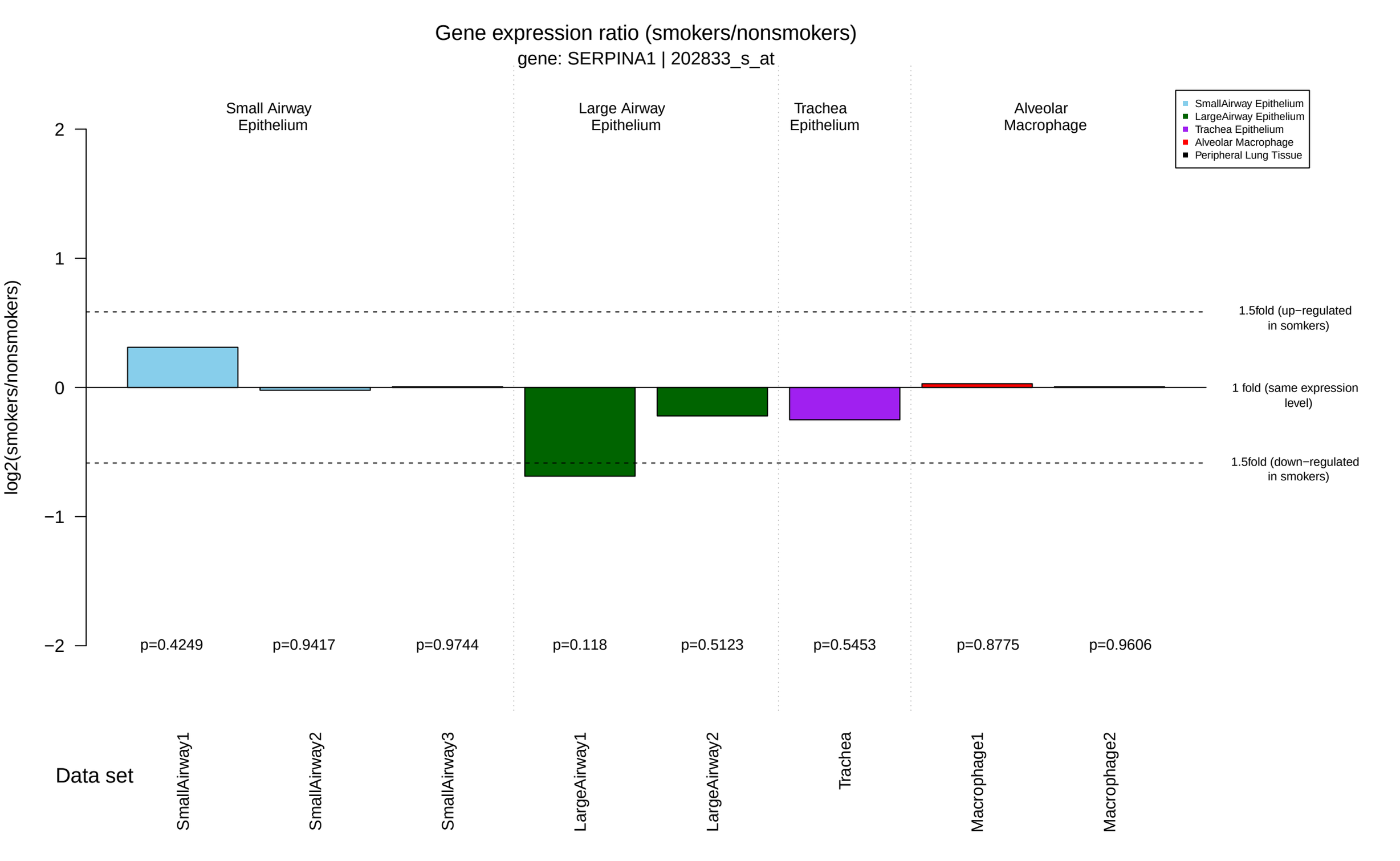
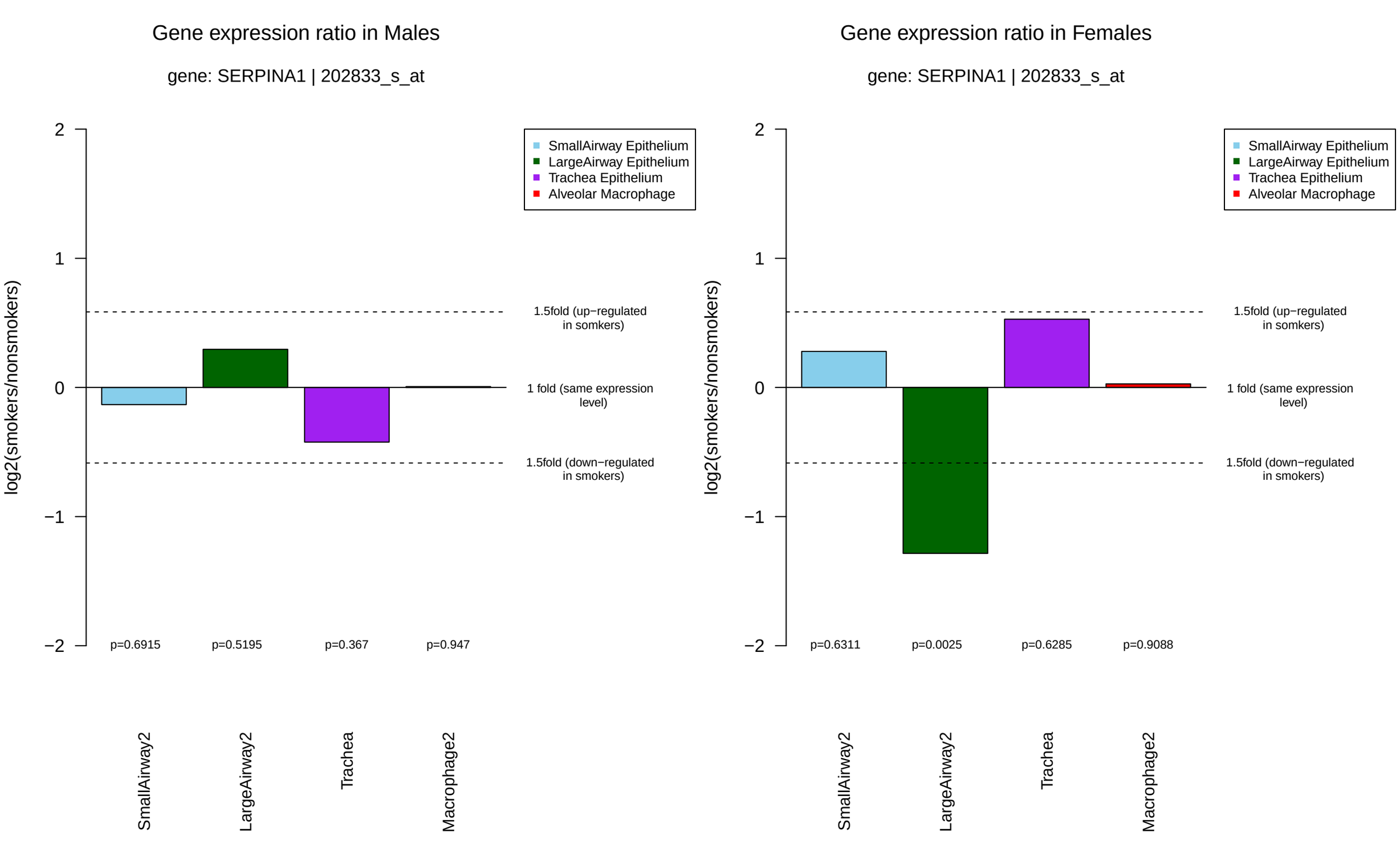
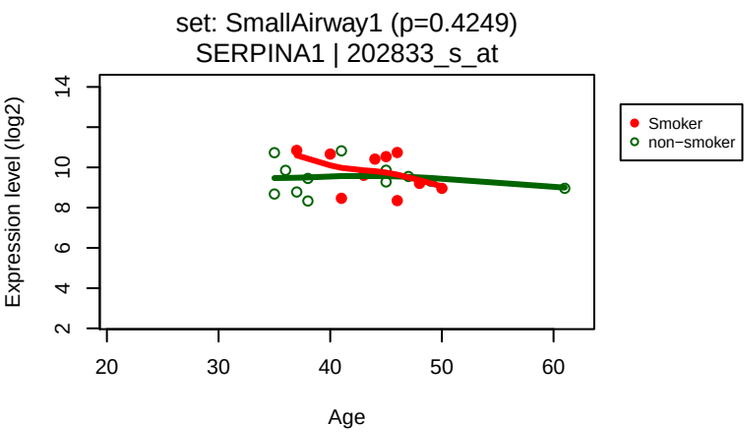
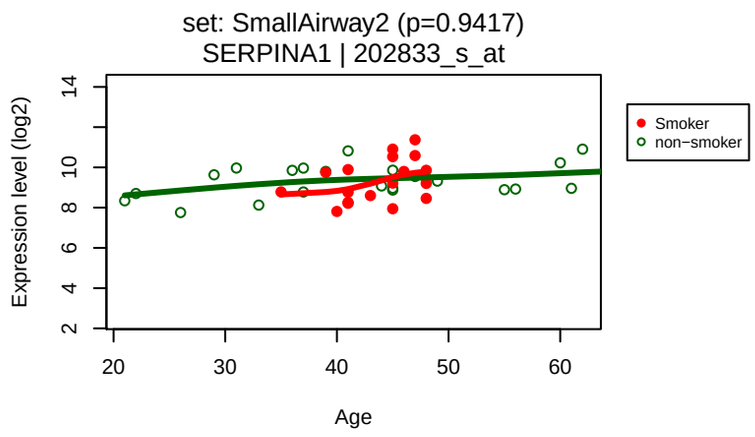
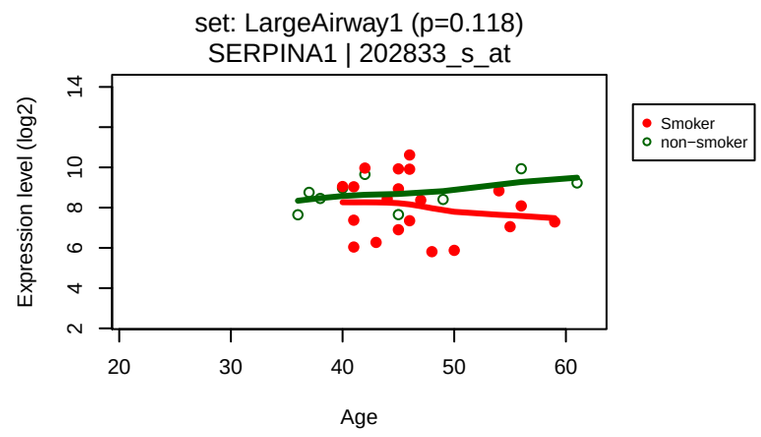

Gene expression of smokers and nonsmokers
gene: SERPINA1 | 202833_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4249
p=0.9417
p=0.9744
p=0.118
p=0.5123
p=0.5453
p=0.8775
p=0.9606
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

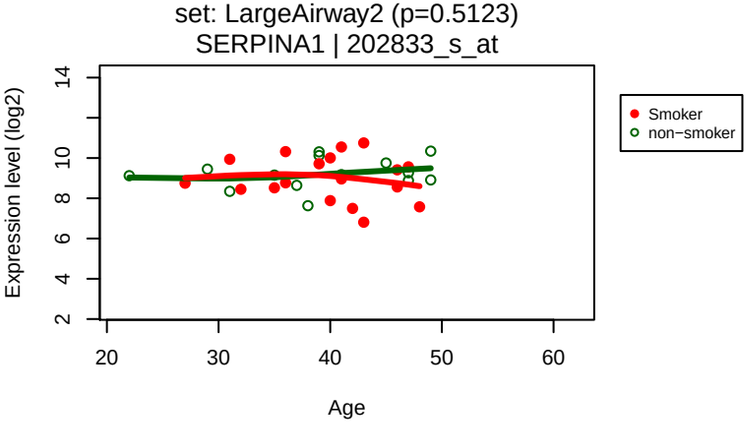

Gene expression ratio (smokers/nonsmokers)
gene: SERPINA1 | 202833_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4249
p=0.9417
p=0.9744
p=0.118
p=0.5123
p=0.5453
p=0.8775
p=0.9606
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

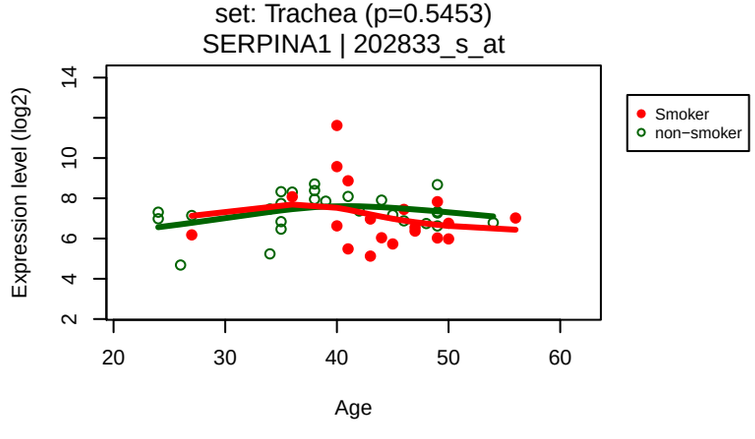

Gene expression ratio in Males
gene: SERPINA1 | 202833_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6915
p=0.5195
p=0.367
p=0.947
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINA1 | 202833_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6311
p=0.0025
p=0.6285
p=0.9088
SmallAirway2
LargeAirway2
Trachea
Macrophage2

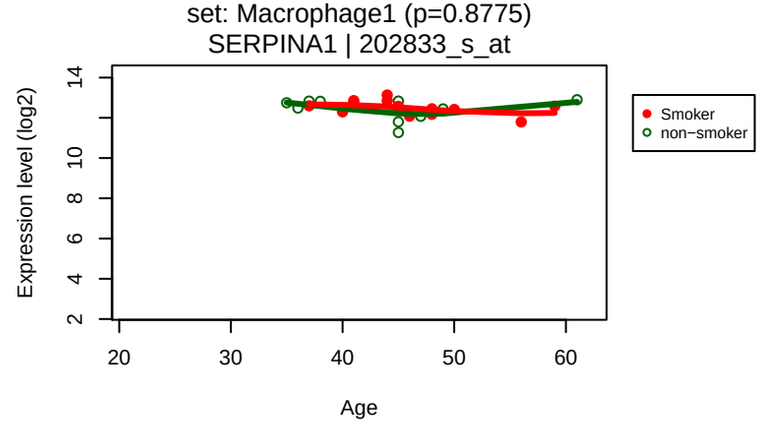

set: SmallAirway1 (p=0.4249)
SERPINA1 | 202833_s_at
Expression level (log2)
Age
Smoker
non−smoker

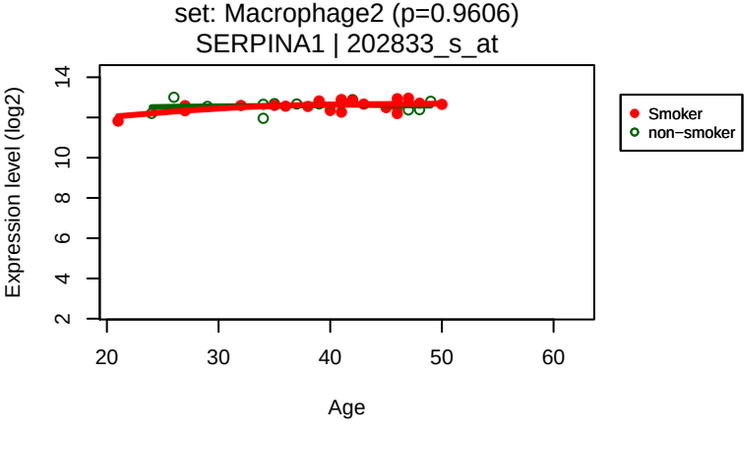

set: SmallAirway2 (p=0.9417)
SERPINA1 | 202833_s_at
Expression level (log2)
Age
Smoker
non−smoker

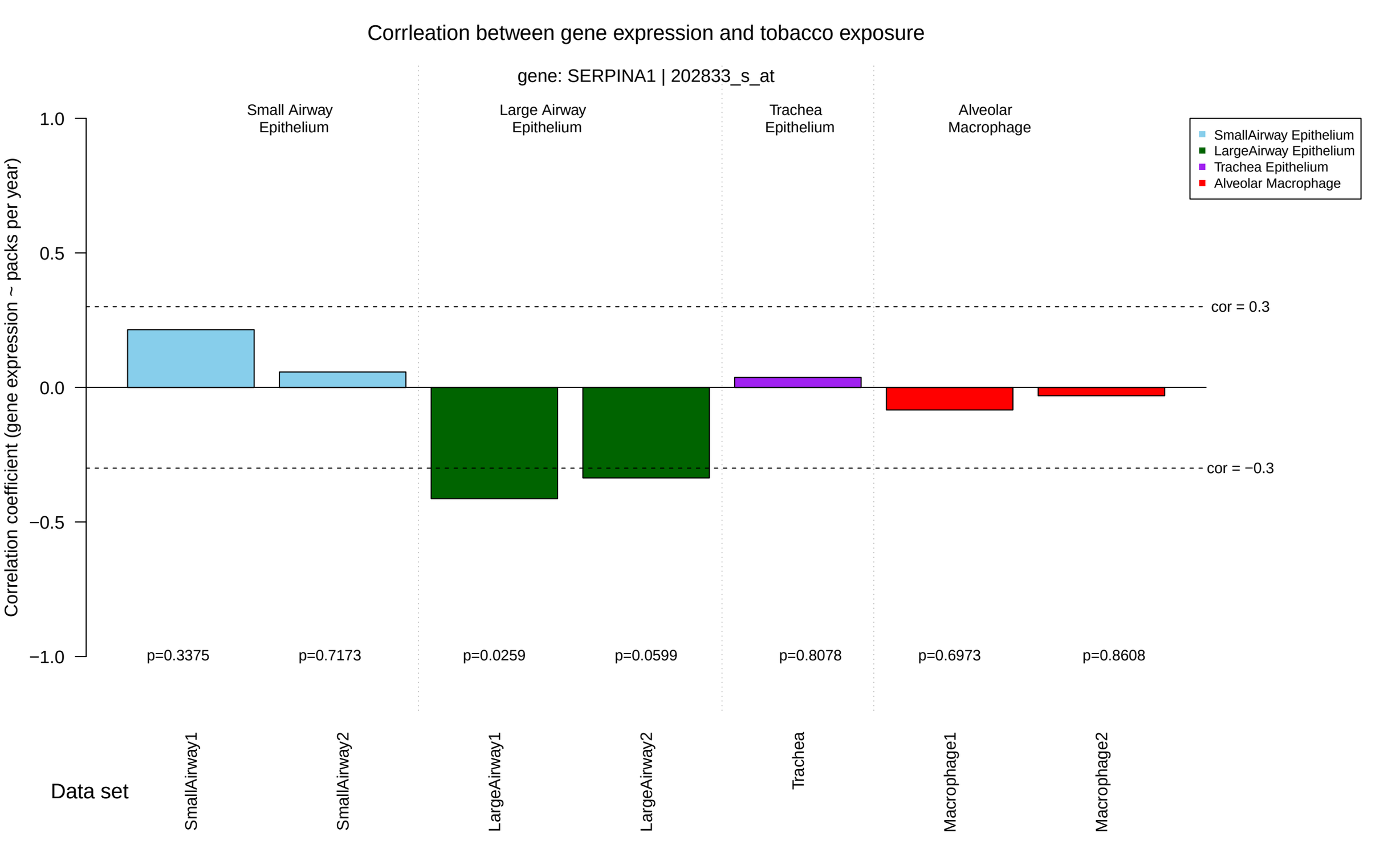

set: LargeAirway1 (p=0.118)
SERPINA1 | 202833_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5123)
SERPINA1 | 202833_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5453)
SERPINA1 | 202833_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8775)
SERPINA1 | 202833_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9606)
SERPINA1 | 202833_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINA1 | 202833_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3375
p=0.7173
p=0.0259
p=0.0599
p=0.8078
p=0.6973
p=0.8608
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2